Metaheuristic-Optimized Machine Learning for Battery State-of-Charge Estimation with Explainable AI

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***Abstract*—**

***Keywords*—Battery state-of-charge, machine learning, genetic algorithm, explainable AI, SHAP**

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III. Dataset and Preprocessing

A. Dataset Description

B. Feature Selection

C. Data Preprocessing

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B. Implementation Details

VI. Results and Discussion

A. Performance Comparison

B. Computational Cost Analysis

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References

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